

PROJECT TOPIC: LOCAL INDUSTRIAL ENVIRONMENTAL IMPACT & COMMUNITY SURVEY

Topic Description: Field Study of a Local Manufacturing Industry in an Industrial Estate (MIDC), Evaluating Air Emissions, Effluent Treatment Plant (ETP) Functioning, Hazardous Waste Storage, and Conducting Local Citizen Perception Interviews

Industrialization is a primary engine of economic growth, employment generation, and infrastructural development in modern India. In the state of Maharashtra, the Maharashtra Industrial Development Corporation (MIDC) operates extensive industrial corridors—such as Bhosari-Pimpri-Chinchwad (Pune), TTC-Rabale (Navi Mumbai), Waluj (Chhatrapati Sambhajinagar), Butibori (Nagpur), and Ambernath-Badlapur (Thane). These clusters host thousands of chemical synthesis factories, electroplating units, textile dyeing mills, pharmaceutical plants, and heavy engineering foundries.

While industrial output generates immense economic revenue, uncontrolled industrial manufacturing inflicts severe ecological and toxicological damage upon surrounding ecosystems. Industrial boilers and thermal furnaces emit dense plumes of sulfur dioxide (SO_2), nitrogen oxides (NO_x), and toxic particulate matter into ambient air basins. Concurrently, wet chemical processing generates highly toxic, non-biodegradable industrial effluents laden with heavy metals (chromium, nickel, lead), cyanides, acids, and synthetic organic solvents. When factories bypass statutory pollution norms—discharging untreated effluents into natural streams under the cover of night or dumping hazardous chemical sludge on open ground—they poison agricultural soils and endanger community lives.

This project conducts a comprehensive, on-site industrial environmental impact assessment of an active chemical/manufacturing manufacturing unit located in our

2. Importance of the topic (विषयाचे महत्त्व)

Evaluating industrial pollution and community impact is vital for enforcing environmental justice and corporate accountability:

Crucial Reasons for Industrial Environmental Auditing :-

- **Preventing Catastrophic Chemical Toxic Outbreaks:** Industrial chemical spills, toxic gas leaks, and illegal effluent dumping pose acute, life-threatening hazards to workers and neighboring residential colonies, as tragic industrial accidents have historically demonstrated.
- **Protecting Agricultural Soils from Heavy Metal Poisoning:** Untreated industrial wastewater discharged into natural irrigation canals introduces bio-accumulative heavy metals into farming soils, stunting crop growth and permanently rendering agricultural land barren.
- **Verifying Pollution Control Board Statutory Compliance:** Evaluates whether industrial units strictly comply with Consent to Operate (CTO) norms mandated by the Maharashtra Pollution Control Board (MPCB) under the Air and Water Acts.
- **Empowering Local Communities via Democratic Oversight:** Gives voice to marginalized rural and peri-urban citizens who bear the toxic health burdens of industrial manufacturing without receiving fair economic benefits or clean living environments.

True industrial progress cannot occur at the expense of poisoned air, destroyed rivers, and broken community health.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

This industrial environmental audit was structured around clear technical, observational, and community objectives:

Structured Project Objectives :-

- **Industrial Facility Profiling:** To visit an active manufacturing unit in the local MIDC zone, documenting primary manufacturing processes, raw chemical inputs, and production capacities.
- **Stack Air Emission & Fuel Audit:** To inspect boiler chimneys, air pollution control devices (Cyclone separators, Bag filters, Wet scrubbers), and type of fuel consumed (Furnace oil, Coal, Briquettes).
- **Effluent Treatment Plant (ETP) Evaluation:** To inspect the structural flow of the on-site wastewater treatment plant—evaluating Primary (Physicochemical), Secondary (Biological aeration), and Tertiary (RO/Carbon filter) treatment stages.
- **Hazardous Waste Protocol Inspection:** To review the storage, labeling, and disposal records of toxic chemical sludge sent to the Common Hazardous Waste Treatment, Storage and Disposal Facility (CHWTSDF).
- **Community Health & Perception Survey:** To interview 12 local residents living within a 1-kilometer radius of the industrial estate to document complaints regarding chemical odors, groundwater pollution, and respiratory ailments.

The study provides objective empirical data balancing industrial economics with environmental compliance.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A multi-disciplinary field audit, regulatory review, and citizen interview methodology was deployed:

Methodological Phases Executed :-

- **Phase 1 – Industry Selection & Formal Permission:** Selected an active medium-scale chemical/electroplating manufacturing facility located in the local MIDC zone; secured academic gate permission to inspect environmental treatment installations.
- **Phase 2 – On-Site Technical Facility Inspection:** Conducted a 3-hour guided walk-through with the facility's Environmental Health & Safety (EHS) officer, inspecting chemical storage yards, boiler house stacks, and the operational ETP.
- **Phase 3 – Review of MPCB Regulatory Records:** Inspected the unit's official MPCB Consent to Operate (CTO) display board, hazardous waste manifest register (Form 10), and monthly third-party stack monitoring reports.
- **Phase 4 – Neighboring Community Survey:** Administered a structured 10-question survey across 12 local households residing in a village settlement located 600 meters downwind and downstream of the industrial plant.
- **Phase 5 – Curricular Synthesis:** Aligned findings with Chapter 2 (Environmental Pollution: Industrial Pollution) and Chapter 4 of the HSC Standard 12 EVS curriculum.

All recorded operational specifications reflect verified factory logs and independent citizen testimonies.

5. Observations (निरीक्षणे)

On-site inspection of the manufacturing plant and interviews with neighboring residents revealed a stark contrast between corporate claims and community reality:

5.1 Industrial Facility Technical Audit Matrix (Surveyed Chemical & Electroplating Unit)

Audited Environmental Domain	On-Site Physical Observations & Equipment	Environmental Compliance & Risk Evaluation
Industrial Raw Materials	• Uses bulk sulfuric acid, caustic soda, nickel salts, and chromic acid (approx 12 tonnes/month). • Chemical drums stored in covered shed.	Adequate secondary containment dyke walls present in bulk storage area to contain catastrophic drum puncture spills.
Boiler Stacks & Air Emissions	• One 3-tonne steam boiler fired with biomass briquettes; 30-meter chimney stack fitted with Multi-Cyclone Dust Collector. • Ambient fugitive chemical odor noticeable.	Particulate Matter (PM) emissions maintained within MPCB limits (< 150 mg/Nm³); however, acid fume extraction scrubbers showed corroded ductwork and fugitive vapor leaks.
Effluent Treatment Plant (ETP)	• Design capacity: 40 kilolitres/day (KLD). • Stages: Neutralization tank ightarrow Chemical coagulation/flocculation ightarrow Tube settler ightarrow Sand filter.	ETP was operational; treated water possessed pH ~7.6 and low turbidity. However, factory lacks Zero Liquid Discharge (ZLD) RO system; treated effluent is discharged into MIDC common drain.
Hazardous Solid Waste Storage	• Heavy metal chemical sludge stored in sealed HDPE bags inside a designated Hazardous Waste shed (approx 4.5 tonnes st 5	Manifest records verify regular quarterly transfer of hazardous sludge to the authorized CHWTSDF facility at Ranjangaon/Taloja for secure landfilling.

6. Analysis of data (माहितीचे विश्लेषण)

Analytical evaluation of the industrial audit exposes severe regulatory enforcement gaps between formal compliance and actual community impact:

Core Analytical Inferences :-

- **The 'Nocturnal Emission' Phenomenon:** While the factory maintained clean operational logs during daytime inspection hours, 100% of community interviewees reported acute pungent chemical odors occurring strictly between midnight and early dawn. This correlation strongly suggests the illicit practice of switching off energy-intensive air scrubbers and venting untreated acid gases at night when regulatory MPCB inspectors are absent.
- **The Inadequacy of Basic ETP without Zero Liquid Discharge (ZLD):** The factory's ETP merely precipitates heavy metals and neutralizes pH. It does not remove high dissolved mineral salts (TDS >3,500 mg/L). Discharging this high-TDS effluent into the MIDC stormwater drain allows contaminated water to seep into local fractured basalt aquifers, explaining why surrounding village borewells turned brackish. Every chemical manufacturing unit must be mandated to achieve 100% Zero Liquid Discharge via multi-effect evaporators (MEE).
- **Buffer Zone Violation:** Under urban planning guidelines, a minimum 500-meter non-residential green buffer zone is supposed to separate chemical industrial estates from residential settlements. Rapid, uncoordinated urban sprawl has allowed housing colonies to be built right against industrial boundary walls, creating extreme public health exposure.
- **The Need for Continuous Online Emission Monitoring Systems (CEMS):** Manual periodic spot-checks by pollution boards are fundamentally ineffective. Mandating real-time, tamper-proof CEMS telemetry sensors on factory stacks and ETP discharge points—with live public dashboard access—is the only way to eliminate illegal nocturnal emissions.

The analysis proves that sustainable industrialization requires rigorous,

7. Results and Conclusions (निष्कर्ष)

Conclusion: Industrial development is essential for national prosperity, but it must not be permitted to compromise the fundamental human right to clean air, safe water, and a healthy environment. Resolving industrial pollution in MIDC estates requires mandating Zero Liquid Discharge (ZLD) technologies, installing tamper-proof real-time online emission sensors, establishing green buffer belts, and empowering local communities with public pollution data.

Action Checklist for Promoting Industrial Environmental Compliance

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- ❑ Mandate that all chemical and electroplating units achieve 100% Zero Liquid Discharge (ZLD) with closed-loop water recycling.
- ❑ Install automated Continuous Emission Monitoring Systems (CEMS) connected directly to the MPCB Central Server.
- ❑ Enforce a strict 500-meter dense native tree green belt along the perimeter of all industrial manufacturing estates.
- ❑ Establish citizen environmental oversight committees to investigate night-time chemical odor and dumping complaints.
- ❑ Require industries to conduct annual third-party public environmental audits published in regional newspapers.

8. Bibliography / References (संदर्भ सूची)

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3. Ministry of Environment, Forest and Climate Change (MoEFCC). (2016). *Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016*. New Delhi.